

Math 777 - Measure theoretic probability

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These notes, adapted from "Probability with Martingales" by David Williams, serve as an introduction to probability through the lens of measure theory. The concepts and understanding of Measure theory will be laid out in the review section of this document, but ideally a course in the subject would be best. These notes also serve to mirror the material taught in [this](#) course from the University of Cambridge by Professor James Norris in [M21](#). These notes were taken in a Independent Study at Hunter College conducted by Professor [Vincent Martinez](#).

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1 Foundations

1.1 Measure Theory

To start off, let's go through some measure theory as a refresher for those of us who have taken that course in another life.

Definition 1. (σ -algebra). Let S be a set. A subset Σ of the power set 2^S is called a σ -algebra (or σ -field) if

- (i) $\emptyset \in \Sigma$
- (ii) If $A \in \Sigma$, then $A^C = S \setminus A \in \Sigma$
- (iii) If $A_1, A_2, \dots \in \Sigma$, then $\bigcup_{n=1}^{\infty} A_n \in \Sigma$

Notation 1. A sigma algebra in our sense is a true algebra, namely the fact that $\cap =$ product, and the sum is defined to be $(A \cup B) \setminus (A \cap B)$

Definition 2. (Measurable space). A *measurable space* is a set equipped with a σ -algebra (S, Σ)

Definition 3. (Borel σ -algebra). Let E be a topological space with topology $\mathcal{T}$. Then the *Borel σ -algebra* $\mathcal{B}(E)$ on E is the σ -algebra generated by $\mathcal{T}$, i.e. the smallest σ -algebra containing $\mathcal{T}$. To put in general terms, $\mathcal{B}(E) = \sigma(\text{open sets})$.

We are often going to look at the Borel σ -algebra on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, and we will just write it as $\mathcal{B}$. We'll primarily be using this when our set S is $\mathbb{R}$ or $\mathbb{R}^+$

Definition 4. (Measure). A function $\mu : \Sigma \rightarrow [0, \infty]$ is a *measure* if

- $\mu(\emptyset) = 0$
- If $A_1, A_2, \dots \in \Sigma$ are disjoint, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

This last property is often described as being countably additive.

Definition 5. (Measure space). A *measure space* is a measurable space with a measure denoted as (S, Σ, μ) . A measure space is called:

- finite if $\mu(S) < \infty$
- σ -finite if there is a sequence $(S_n : n \in \mathbb{N})$ of elements of Σ s.t.

$$\mu(S_n) < \infty (\forall n \in \mathbb{N}) \text{ and } \bigcup S_n = S$$

We can use our better judgement when talking about finite measures, and generally the same for σ -finite measures, however once it's no longer σ -finite, it can be crazy.

Definition 6. (Measurable function). Let (S_1, Σ_1) and (S_2, Σ_2) be measurable spaces. Then $f : S_1 \rightarrow S_2$ is said to be *measurable* if $A \in \Sigma_2$ implies $f^{-1}(A) \in \Sigma_1$.

Notation 2. For (S, Σ) a measurable space, we write $m\Sigma$ for the set of measurable functions $S \rightarrow \mathbb{R}$.

We write $m\Sigma^+$ to be the positive measurable functions, which are allowed to take value ∞ .

Note that we do *not* allow taking values $\pm\infty$ in the first case.

Theorem 1. Let (S, Σ, μ) be a measure space. Then there exists a unique function $\tilde{\mu} : m\Sigma^+ \rightarrow [0, 1]$ satisfying

- (a) $\tilde{\mu}(1_A) = \mu(A)$, where 1_A is the indicator function of A .
- (b) Linearity: $\tilde{\mu}(\alpha f + \beta g) = \alpha \tilde{\mu}(f) + \beta \tilde{\mu}(g)$ if $\alpha, \beta \in \mathbb{R}_{\geq 0}$ and $f, g \in m\Sigma^+$
- (c) Monotone convergence: $\tilde{\mu}(f_n) \rightarrow \tilde{\mu}(f)$ as $n \rightarrow \infty$ whenever $(f_n : n \in \mathbb{N})$ is a non-decreasing sequence in $m\Sigma^+$ with pointwise limit f

We call $\tilde{\mu}$ the *integral* with respect to μ , and we will write it as μ from now on...

Definition 7. (Simple function). A function f is *simple* if there exists $\alpha_n \in \mathbb{R}_{\geq 0}$ and $A_n \in \Sigma$ for $1 \leq n \leq k$ such that

$$f = \sum_{n=1}^k \alpha_n \mathbb{1}_{A_n}$$

a.k.a. It is a finite linear combination of indicator functions of measurable sets, with positive coefficients. From the first two properties of a measure, we can observe that the integral of a simple function must be:

$$\mu(f) = \sum_{n=1}^k \alpha_n \mu(A_n)$$

One convenient observation is that a function is simple iff it takes on only finitely many values. We then see that if $f \in m\Sigma^+$, then

$$f_n = 2^{-n} \lfloor 2^n f \rfloor \wedge n$$

is the sequence of simple functions approximating f from below. Thus, given monotone convergence, this shows:

$$\mu(f) = \lim \mu(f_n)$$

and this proves the uniqueness part of the theorem.

Definition 8. (Almost everywhere). We say that $f = g$ almost everywhere if

$$\mu(\{x \in S : f(x) \neq g(x)\}) = 0$$

We say f is a *version* of g . This element x that breaks the equality is said to be μ -null

Example 8.1. Let $l_n = 1_{[n, n+1]}$. Then $\mu(l_n) = 1$ for all n , but also $f_n \rightarrow 0$ and $\mu(0) = 0$. So the "monotone" part of monotone convergence is important.

So if the sequence is not monotone, then the measure does not preserve limits, but it turns out we still have an inequality.

Lemma 1. (Fatou's lemma). Let $f_i \in m\Sigma^+$. Then

$$\mu(\liminf_n f_n) \leq \liminf_n \mu(f_n)$$

The proof is by applying the monotone convergence theorem to the non-decreasing sequence of functions $(\inf_{m \geq n} f_m : n \in \mathbb{N})$.

(Reverse Fatou's lemma).

Definition 9. (Integratable function). We say a function $f \in m\Sigma$ is *integrable* if $\mu(|f|) < \infty$. We write $L^1(S)$ or just L^1 for the space of integrable functions. We extend μ to L^1 by

$$\mu(f) = \mu(f^+) - \mu(f^-)$$

where $f^\pm = (\pm f) \wedge 0$.

Theorem 2. (Dominated convergence theorem). If $f_i \in m\Sigma$ and $f_i \rightarrow f$ a.e., such that there exists $g \in L^1$ s.t. $|f_i| \leq g$ a.e. Then

$$\mu(f) = \lim \mu(f_n)$$

Proof Apply Fatou's lemma to $g - f_n$ and $g + f_n$

Definition 10. (Product σ -algebra). Let (S_1, Σ_1, μ_1) and (S_2, Σ_2, μ_2) be finite measure spaces. Then the product σ -algebra $\Sigma_1 \otimes \Sigma_2$ is the smallest σ -algebra on $E_1 \times E_2$ containing all sets of the form $A_1 \times A_2$, where $A_i \in \Sigma_i$

Theorem 3. If (E_1, Σ_1, μ_1) and (E_2, Σ_2, μ_2) are σ -finite measure spaces, then there exists a *unique* measure μ on $\Sigma_1 \otimes \Sigma_2$ where

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$$

for all $A_i \in \Sigma_i$. This is called the *product measure*.

Theorem 4. (Fubini's/Tonelli's theorem). If $f = f(x_1, x_2) \in m\Sigma^+$ with $\Sigma = \Sigma_1 \otimes \Sigma_2$, then the functions

$$x_1 \mapsto \int f(x_1, x_2) d\mu_2(x_2) \in m\Sigma_1^+$$

$$x_2 \rightarrow \int f(x_1, x_2) d\mu_1(x_1) \in m\Sigma_2^+$$

and

$$\begin{aligned} \int_E f du &= \int_{E_1} \left(\int_{E_2} f(x_1, x_2) d\mu_2(x_2) \right) d\mu_1(x_1) \\ &= \int_{E_2} \left(\int_{E_1} f(x_1, x_2) d\mu_1(x_1) \right) d\mu_2(x_2) \end{aligned}$$

1.2 Conditional expectation

We say that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space if it is a measure space with the property that $\mathbb{P}(\Omega) = 1$. We are now going to focus on probability theory, which, mathematically, just means we assume $\mu(E) = 1$.

In statistics, we use the notion of:

- $S = \Omega$ (Set $\rightarrow$ Sample Space)
- $\Sigma = \mathcal{F}$ (sigma algebra on $E \rightarrow$ event space on Ω)
- $\mu = \mathbb{P}$ (measure $\rightarrow$ probability measure/function)
- $\int d\mu = \mathbb{E}$ (integral of a random variable with respect to $\mathbb{P}$ is the expectation)
- measurable sets $\in \Sigma \rightarrow$ events $\in \mathcal{F}$
- $f = X$ (Measurable function on $(E, \Sigma) \rightarrow$ Random variable on $(\Omega, \mathcal{F})$)
- element $\in E \rightarrow \omega$ /sample point/realization $\in \Omega$

In the context of probability theory, we consider almost surely to mean almost everywhere.

Suppose $B \in \mathcal{F}$, with $\mathbb{P}(B) > 0$. Then the conditional probability of the event A given B is

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

This means, what is the probability of A happening, given that B has occurred. Since we assume B has occurred, we should restrict to the subset of the probability space where B has happened. To make this a probability space, we scale the probability measure $\mathbb{P}(B)$. Then given any event A , we take the probability of $A \cap B$ under this probability measure.

To generalize, if X is a random variable, the conditional expectation of X given B is just the expectation under this new probability measure,

$$\mathbb{E}[X|B] = \frac{\mathbb{E}[X\mathbb{1}_B]}{\mathbb{P}[B]}.$$

We've all seen this before in elementary school, and it's not exactly anything new. One natural generalization would be to allow B to vary.

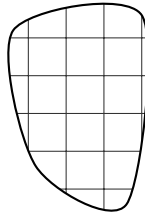
Let $G_1, G_2, \dots \in \mathcal{F}$ be disjoint events s.t. $\bigcup_n G_n = \Omega$. Let

$$\mathcal{G} = \sigma(G_1, G_2, \dots) = \left\{ \bigcup_{n \in I} G_n : I \subseteq \mathbb{N} \right\}$$

Let $X \in L^1$. We then define

$$Y = \sum_{n=1}^{\infty} \mathbb{E}(X|G_n) \mathbb{1}_{G_n}$$

To visualize what's happening here, suppose a random outcome ω happens. To compute Y , we figure out which of the G_n our ω belongs to. Let's say $\omega \in G_k$, then Y will return the expected value of X given that we live in G_k . All that matters is which G_n the outcome belongs to. We can visualize G_n as cutting up the sample space Ω into slices.



We then average out X in each of these compartments to obtain Y . This is called the conditional expectation of X given $\mathcal{G}$, written as $\mathbb{E}(X|\mathcal{G})$.

Lemma 2. The conditional expectation $Y = \mathbb{E}(X|\mathcal{G})$ satisfies the following properties:

- Y is $\mathcal{G}$ -measurable
- We have $Y \in L^1$, and

$$\mathbb{E}Y\mathbb{1}_A = \mathbb{E}X\mathbb{1}_A$$

for all $A \in \mathcal{G}$

Proof. It is clear that Y is $\mathcal{G}$ -measurable. To show that it is L^1 , we compute

$$\begin{aligned}
 \mathbb{E}[|Y|] &= \mathbb{E} \left| \sum_{n=1}^{\infty} \mathbb{E}(X|G_n) \mathbb{1}_{G_n} \right| \\
 &\leq \mathbb{E} \sum_{n=1}^{\infty} \mathbb{E}(|X| | G_n) \mathbb{1}_{G_n} \\
 &= \sum \mathbb{E}(\mathbb{E}(|X| | G_n) \mathbb{1}_{G_n}) \\
 &= \sum \mathbb{E}|X| \mathbb{1}_{G_n} \\
 &= \mathbb{E} \sum |X| \mathbb{1}_{G_n} \\
 &= \mathbb{E}|X| \\
 &< \infty
 \end{aligned}$$

where the monotone convergence theorem allows us to swap the expectation and the sum. In the final part we can enumerate the elements in $\mathcal{G}$ to see that they all satisfy the last property.

It turns out that for any σ -subalgebra $\mathcal{G} \subseteq \mathcal{F}$, we can construct the conditional expectation $\mathbb{E}(X|\mathcal{G})$, given by the two properties above.

Theorem 5. (Existence and uniqueness of conditional expectation). Let $X \in L^1$, and $\mathcal{G} \subseteq \mathcal{F}$. Then there exists a random variable Y s.t.

- Y is $\mathcal{G}$ -measurable
- $Y \in L^1$, and $\mathbb{E}X\mathbb{1}_A = \mathbb{E}Y\mathbb{1}_A$, for all $A \in \mathcal{G}$

Moreover, if Y' is another random variable satisfying these conditions, then $Y' = Y$ a.s.

We will call Y a (version of) the conditional expectation given $\mathbb{E}$

We will denote the conditional expectation of a variable X from condition $\mathcal{G}$ as $\mathbb{E}(X|\mathcal{G})$, and if $X = \mathbb{1}_A$, we will write $\mathbb{P}(A|\mathcal{G}) = \mathbb{E}(\mathbb{1}_A|\mathcal{G})$

Since Z is a random variable, then $\sigma(Z) = \{Z^{-1}(B) : B \in \mathcal{B}\}$. In this case, we will denote $\mathbb{E}(X|Z)$ as $\mathbb{E}(X|\sigma(Z))$

By bounded convergence, it stands that $\mathbb{E}XZ = \mathbb{E}YZ$ for all bounded $\mathcal{G}$ -measurable functions Z .

Proof. Consider the case where $X \in L^2(\Omega, \mathcal{F}, \mu)$. Then we know that for any $\mathcal{G} \in \mathcal{F}$, the space $L^2(\mathcal{G})$ is a hilbert space where

$$\langle X, Y \rangle = \mu(XY)$$

Since $L^2(\mathcal{G})$ is a closed subspace of $L^2(\mathcal{F})$, we can define Y to be the orthogonal projection of X onto $L^2(\mathcal{G})$. This means Y is $\mathcal{G}$ -measurable and by extension, $X - Y$ is orthogonal to $L^2(\mathcal{G})$. So $\mathbb{E}(X - Y)Z = 0 \forall Z \in L^2(\mathcal{G})$. In particular, since the measure space is finite, the indicator function of any measurable subset is in L^2 which proves existence.

We next focus on the case where $X \in m\Sigma^2$. Let

$$X_n = X \wedge n$$

We want to use the monotone convergence theorem to obtain the result, to do so:

Claim If (X, Y) and (X', Y') satisfy the conditions of the theorem, and $X' \geq X$ a.s. then $Y' \geq Y$ a.s.

Proof Define the event $A = \{Y' \leq Y\} \in \mathcal{G}$. Consider the event $Z = (Y - Y')\mathbb{1}_A$. Then $Z \geq 0$. We then have

$$\mathbb{E}Y'\mathbb{1}_A = \mathbb{E}X'\mathbb{1}_A \geq \mathbb{E}X\mathbb{1}_A = \mathbb{E}Y\mathbb{1}_A$$

So it follows that we also have $\mathbb{E}(Y - Y')\mathbb{1}_A \leq 0$. So in fact $\mathbb{E}Z = 0$. So $Y' \geq Y$ a.s.

We can now define $Y_n = \mathbb{E}(X_n|\mathcal{G})$, picking them so that $\{Y_n\}$ is increasing. We can then take $Y_\infty = \lim Y_n$. Then Y_∞ is $\mathcal{G}$ -measurable, and by monotone convergence, if $A \in \mathcal{G}$, then

$$\mathbb{E}X\mathbb{1}_A = \lim \mathbb{E}X_n\mathbb{1}_A = \lim \mathbb{E}Y_n\mathbb{1}_A = \mathbb{E}Y_\infty\mathbb{1}_A$$

Now if $\mathbb{E}X < \infty$, then $\mathbb{E}Y_\infty = \mathbb{E}X < \infty$. So we know that Y_∞ is finite a.s. and we can define $Y = Y_\infty\mathbb{1}_{Y_\infty < \infty}$.

Finally, if we say $X \in L^1$ and $X = X^+ - X^-$. We can then define $Y^\pm = \mathbb{E}(X^\pm|\mathcal{G})$, and take $Y = Y^+ - Y^-$. This suffices for uniqueness.

Lemma 3. If Y is $\sigma(Z)$ -measurable, then there exists some $h : \mathbb{R} \rightarrow \mathbb{R}$ borel measurable s.t. $Y = h(Z)$. In particular,

$$\mathbb{E}(X|Z) = h(Z) \text{ a.s.}$$

for some $h : \mathbb{R} \rightarrow \mathbb{R}$.

We can then define $\mathbb{E}(X|Z = z) = h(z)$. This allows for $\mathbb{P}(Z = z) = 0$

Example 10.1. Let $(U, V) \in \mathbb{R}^2$ with density $f_{U,V}(u, v)$, so that for any $B_1, B_2 \in \mathcal{B}$, we have

$$\mathbb{P}(U \in B_1, V \in B_2) = \int_{B_1} \int_{B_2} f_{U,V}(u, v) du dv$$

We want to compute $\mathbb{E}(h(V)|U)$, where $h : \mathbb{R} \rightarrow \mathbb{R}$ is borel measurable. We can define

$$f_U(u) = \int_{\mathbb{R}} f_{U,V}(u, v) dv,$$

and we define the conditional density of V given U to be

$$F_{|U}(v|u) = \frac{f_{U,V}(u, v)}{f_U(u)}$$

We define

$$g(u) = \int h(v) f_{V|U}(v|u) dv$$

We claim that $\mathbb{E}(h(V)|U)$ is just $g(U)$.

We can check this by verifying the following:

1. It is clearly $\sigma(U)$ -measurable.
2. Fix $A \in \sigma(U)$. Then $A = \{(u, v) : u \in B\}$ for some B . Then

$$\begin{aligned} \mathbb{E}[h(V) \mathbb{1}_A] &= \int \int h(v) \mathbb{1}_{u \in B} f_{U,V}(u, v) du dv \\ &= \int \int h(v) \mathbb{1}_{u \in B} f_{V|U}(v|u) du dv \\ &= \int g(U) \mathbb{1}_{u \in B} f_U(u) du \\ &= \mathbb{E}(g(U) \mathbb{1}_A) \end{aligned}$$

as desired.

This demonstrates that to compute conditional expectation, we can use our intuition to guess what conditional expectations should be, and then check that it satisfies the two characterizing properties.

Some immediate properties of conditional expectation.

- (i) $\mathbb{E}(X|\mathcal{G}) = X \iff X$ is $\mathcal{G}$ -measurable
- (ii) $\mathbb{E}(\mathbb{E}(X|\mathcal{G})) = \mathbb{E}X$

(iii) If $X \geq 0$ a.s., then $\mathbb{E}(X|\mathcal{G}) \geq 0$

(iv) If X and $\mathcal{G}$ are independent, then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$

(v) If $a, b \in \mathbb{R}$, and $X_1, X_2 \in L^1$, then

$$\mathbb{E}(aX_1 + bX_2|\mathcal{G}) = a\mathbb{E}(X_1|\mathcal{G}) + b\mathbb{E}(X_2|\mathcal{G})$$

(vi) $\text{spz } X_n \nearrow X$. Then

$$\mathbb{E}(X_n|\mathcal{G}) \nearrow \mathbb{E}(X|\mathcal{G})$$

(vii) *Fatou's lemma*: If X_n are non negative measurable, then

$$\mathbb{E}(\liminf_{n \rightarrow \infty} X_n|\mathcal{G}) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n|\mathcal{G})$$

(viii) *Dominated convergence theorem*: If $X_n \rightarrow X$ and $Y \in L^1$ s.t. $Y \geq |X_n|$ for all n , then

$$\mathbb{E}(X_n|\mathcal{G}) \rightarrow \mathbb{E}(X|\mathcal{G})$$

(ix) *Jensen's inequality*: If $c : \mathbb{R} \rightarrow \mathbb{R}$ is convex, then

$$\mathbb{E}(c(X)|\mathcal{G}) \geq c(\mathbb{E}(X)|\mathcal{G})$$

(x) *Tower property*: If $\mathcal{H} \subseteq \mathcal{G}$, then

$$\mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{H}) = \mathbb{E}(X|\mathcal{H})$$

(xi) For $p \geq 1$,

$$\|\mathbb{E}(X|\mathcal{G})\|_p \leq \|X\|_p$$

(xii) If Z is bounded and $\mathcal{G}$ -measurable, then

$$\mathbb{E}(ZX|\mathcal{G}) = Z\mathbb{E}(X|\mathcal{G})$$

(xiii) Let $X \in L^1$ and $\mathcal{G}, \mathcal{H} \subseteq \mathcal{F}$. Assume that $\sigma(X, \mathcal{G})$ is independent of $\mathcal{H}$. Then

$$\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X|\sigma(\mathcal{G}, \mathcal{H}))$$

Proof

(i) Clear

(ii) Let $A = \omega$

(iii) Shown in the proof

- (iv) via the property of expected value of independant variables
- (v) RHS satisfies the unique characterizing property of the LHS
- (vi) By construction
- (vii) Same as the unconditional proof and 6
- (viii) Same as the unconditional proof and 7
- (ix) Same as the unconditional proof
- (x) The LHS satisfies the characterizing property of the RHS
- (xi) Using the convexity of $|x|^p$, Jensen's inequality tells us

$$\begin{aligned}
 \|\mathbb{E}(X|\mathcal{G})\|_p^p &= \mathbb{E}|\mathbb{E}(X|\mathcal{G})|^p \\
 &\leq \mathbb{E}(\mathbb{E}(|X|^p|\mathcal{G})) \\
 &= \mathbb{E}|X|^p \\
 &= \|X\|_p^p
 \end{aligned}$$

- (xii) If $Z = \mathbb{1}_B$, and let $b \in \mathcal{G}$. Then

$$\mathbb{E}(Z\mathbb{E}(X|\mathcal{G})\mathbb{1}_A) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})\mathbb{1}_A\mathbb{1}_B) = \mathbb{E}(X\mathbb{1}_A\mathbb{1}_B) = \mathbb{E}(ZX\mathbb{1}_A)$$

So the lemma holds. Linearity then implies the result for Z simple, then apply our favorite convergence theorems.

- (xiii) Take $B \in \mathcal{H}$ and $A \in \mathcal{G}$. Then

$$\begin{aligned}
 \mathbb{E}(\mathbb{E}(X|\sigma(\mathcal{G}, \mathcal{H}))\mathbb{1}_A\mathbb{1}_B) &= \mathbb{E}(X\mathbb{1}_A\mathbb{1}_B) \\
 &= \mathbb{E}(X\mathbb{1}_A)\mathbb{P}(B) \\
 &= \mathbb{E}(\mathbb{E}(X|\mathcal{G})\mathbb{1}_A)\mathbb{P}(B) \\
 &= \mathbb{E}(\mathbb{E}(X|\mathcal{G})\mathbb{1}_A\mathbb{1}_B)
 \end{aligned}$$

Lemma 4. If $X \in L^1$, then the family of RVs $Y_{\mathcal{G}} = \mathbb{E}(X|\mathcal{G})$ for all $\mathcal{G} \subseteq \mathcal{F}$ is uniformly integrable. In other words, for all $\epsilon > 0$, there exists some $\lambda > 0$ such that

$$\mathbb{E}(Y_{\mathcal{G}}\mathbb{1}_{|Y_{\mathcal{G}}|>\lambda}) < \epsilon$$

for all $\mathcal{G}$.

Proof Let $\epsilon > 0$. There exists some $\delta > 0$ s.t. $\mathbb{E}|X|\mathbb{1}_A < \epsilon$ for any A with $\mathbb{P}(A) < \delta$.

Take $Y = \mathbb{E}(X|\mathcal{G})$. Then by Jensen's, we know

$$|Y| \leq \mathbb{E}(|X| | \mathcal{G})$$

In particular, we have

$$\mathbb{E}|Y| \leq \mathbb{E}|X|$$

By Markov's inequality, we have

$$\mathbb{P}(|Y| \geq \lambda) \leq \frac{\mathbb{E}|Y|}{\lambda} \leq \frac{\mathbb{E}|X|}{\lambda}$$

So take λ such that $\frac{\mathbb{E}|X|}{\lambda} < \delta$. So we have

$$\mathbb{E}(|Y| \mathbb{1}_{|Y| \geq \lambda}) \leq \mathbb{E}(\mathbb{E}(|X| | \mathcal{G}) \mathbb{1}_{|Y| \geq \lambda}) = \mathbb{E}(|X| \mathbb{1}_{|Y| \geq \lambda}) < \epsilon$$

Since $\mathbb{1}_{|Y| \geq \lambda}$ is a $\mathcal{G}$ -measurable function.

1.3 Old solution to abracadabra problem

Section 4.9 - If the monkey only types capital letter, and is on every occasion equally likely to type any of the 26, how long on average will it take him to produce the sequence "ABRACADABRA".

This was my initial approach to solving the problem without the use of martingales, expanded reasoning is shown in Chapter 3.

Autocorrelation vector of word w ("ABRACADABRA"):

$$c = (c_0, c_1, \dots, c_{n-1}) \text{ where } c_i = \mathbb{1}_{\text{str}[i:i+1] == \text{str}[n-i-1:]}$$

In abracadabra, the only cases where this occurs is $i=0$ ("a"), $i=3$ ("abra"), and $i=10$ ("abracadabra").

Therefore the autocorrelation vector will be $(1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1)$, using this we can construct the autocorrelation polynomial to be:

$$c(z) = c_0 z^0 + c_1 z^1 + \dots + c_{n-1} z^{n-1} = 1 + z^3 + z^{10}$$

With this, we can use the ordinary generating function (OGF) for a language of words not containing the string:

$$S(z) = \frac{c(z)}{z^k + (1 - z|A|)c(z)}$$

Thus our generating function must be:

$$S(z) = \frac{1 + z^3 + z^{10}}{z^{11} + (1 - 26z)(1 + z^3 + z^{10})}$$

To find the average wait time the first occurrence of a pattern, we can use $z = 1/26$ (since our language has only 26 letters)

$$S(1/26) = \frac{1 + \frac{1}{26^7} \frac{1}{26^{11}}}{\frac{1}{26^{11}} + 0} = \frac{\frac{26^{11} + 26^4 + 26}{26^{11}}}{\frac{1}{26^{11}}} = 26^{11} + 26^4 + 26$$

2 Martingale Theory

2.1 Filtrations

Suppose we'd like to model some random variable that evolves with time. For example, in a simple random walk, X_n we could model the position we are at time n . To do so, we'll want to have some σ -algebra $\mathcal{F}_n$ that tells us the "information we have at time n ". This structure is known as a *filtration*.

Definition 11. (Filtration). A *filtration* is a sequence of σ -algebras $(\mathcal{F}_n)_{n \geq 0}$ such that $\mathcal{F} \supseteq \mathcal{F}_{n+1} \supseteq \mathcal{F}_n$ for all n . We define $\mathcal{F}_\infty = \sigma(\mathcal{F}_0, \mathcal{F}_1, \dots) \subseteq \mathcal{F}$.

We will then assume our probability triple $(\Omega, \mathcal{F}, \mathbb{P})$ is equipped with a filtration $(\mathcal{F}_n)_{n \geq 0}$.

Definition 12. (Stochastic process in discrete time). A *stochastic process* (in discrete time) is a sequence of RVs $(X_n)_{n \geq 0}$. (in most cases we'd like the X_n to interact nicely with our filtration),

Definition 13. (Natural filtration). The *natural filtration* of $(X_n)_{n \geq 0}$ is given by

$$\mathcal{F}_n^X = \sigma(X_1, \dots, X_n)$$

Definition 14. (Adapted process). We say that $(X_n)_{n \geq 0}$ is *adapted* (to $(\mathcal{F}_n)_{n \geq 0}$) if X_n is $\mathcal{F}_n$ -measurable for all $n \geq 0$. Equivalently, if $\mathcal{F}_n^X \subseteq \mathcal{F}_n$.

Definition 15. (Integrable process). A process $(X_n)_{n \geq 0}$ is *integrable* if $X_n \in L^1$ for all $n \geq 0$

We can now describe a martingale!

Definition 16. (Martingale). An integrable adapted process $(X_n)_{n \geq 0}$ is a *martingale* if for all $n \geq m$, we have

$$\mathbb{E}(X_n | \mathcal{F}_m) = X_m$$

We say it's a *super-martingale* if

$$\mathbb{E}(X_n | \mathcal{F}_m) \leq X_m$$

and a *sub-martingale* if

$$\mathbb{E}(X_n | \mathcal{F}_m) \geq X_m$$

But what does this mean conceptually? The idea of a martingale is that we cannot predict whether X_n will go up or down in the future even if we have all the information up to the present. For example, if X_n denotes the wealth of a gambler in a gambling game, then in some sense $(X_n)_{n \geq 0}$ being a martingale means the game is "fair" (in the sense of a fair dice).

Note that $(X_n)_{n \geq 0}$ is a super-martingale iff $(-X_n)_{n \geq 0}$ is a sub-martingale, and if $(X_n)_{n \geq 0}$ is a martingale, then it is both a super-martingale and a sub-martingale. These extra notions allow simplicity in the proof as we can establish a result for a super-martingale (or sub-martingale), and then by applying the results to both $(X_n)_{n \geq 0}$ and $(-X_n)_{n \geq 0}$, we obtain a stronger result that holds for martingales.

2.2 Optimal stopping

The optional stopping theorem gives us a very powerful result that comes from our construction of a martingale. To establish an optional stopping theorem, we'll need the notion of a stopping time.

Definition 17. (Stopping time). A *stopping time* is a RV $T : \Omega \rightarrow \mathbb{N}_{\geq 0} \cup \{\infty\}$ such that

$$\{T \leq n\} \in \mathcal{F}_n$$

for all $n \geq 0$.

This means that at time n , if we want to know if T has occurred, we can determine it by using the information we have at time n .

Note that T is a stopping time iff $\{t = n\} \in \mathcal{F}_n$ for all n , since if T is a stopping time, then

$$\{T = n\} = \{T \leq n\} \setminus \{T \leq n - 1\}$$

and $\{T \leq n - 1\} \in \mathcal{F}_{n-1} \subseteq \mathcal{F}_n$. Conversely,

$$\{T \leq n\} = \bigcup_{k=1}^n \{T = k\} \in \mathcal{F}_n$$

This does not hold in the continuous case.

Example 17.1. If $B \in \mathcal{B}(\mathbb{R})$, then we can define

$$T = \inf\{n : X_n \in B\}$$

Then this is the stopping time.

On the other hand,

$$T = \sup\{n : X_n \in B\}$$

is generally not a stopping time.

Given a stopping time T , we can make the following notations for simplicity:

Definition 18. (X_T). For a stopping time T , we define the RV X_T by

$$X_T(\omega) = X_{T(\omega)}(\omega)$$

on $\{T < \infty\}$, and 0 otherwise.

Later we will see that the limit $X_\infty = \lim_{n \rightarrow \infty} X_n$. We define $X_T(\omega)$ to be $X_\infty(\omega)$ if $T = \infty$. Similarly we can say:

Definition 19. (Stopped process). The *stopped process* is defined by

$$(X_n^T)_{n \geq 0} = (X_{T \wedge n}(\omega))_{n \geq 0}$$

This says that we stop evolving the random variable X once T has occurred.

We'd like to say that X_t is " $\mathcal{F}_T$ -measurable", i.e. to compute X_T , we only need to know if the information up to time T . This allows for our definition of $\mathcal{F}_T$ as follows:

Definition 20. ($\mathcal{F}_t$). For a stopping time T , define

$$\mathcal{F}_t = \{A \in \mathcal{F}_\infty : A \cap \{T \leq n\} \in \mathcal{F}_n\}$$

Which is a σ -algebra

Example 20.1. If $T \equiv n$, is constant, then $\mathcal{F}_T = \mathcal{F}_n$

Some properties of these objects are:

- (i) If $T, S, (T_n)_{n \geq 0}$ are all stopping times, then

$$T \vee S, T \wedge S, \sup_n T_n, \inf_n T_n, \limsup T_n, \liminf T_n$$

are all stopping times

- (ii) $\mathcal{F}_T$ is a σ -algebra
- (iii) $S \leq T$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$
- (iv) $X_t \mathbb{1}_{T < \infty}$ is $\mathcal{F}_t$ -measurable
- (v) If (X_n) is an adapted process, then so is $(X_n^T)_{n \geq 0}$ for any stopping time T
- (vi) If (X_n) is an integrable process, then so is $(X_n^T)_{n \geq 0}$ for any stopping time T

Definition 21. (Optional stopping theorem). Let $(X_n)_{n \geq 0}$ be a super-martingale with $S \leq T$ bounded stopping times. Then

$$\mathbb{E}X_t \leq \mathbb{E}X_S$$

This theorem from a conceptual point of view means that if X is a martingale, then it is both a super-martingale and sub-martingale. So we can apply this to both X and $-X$ to find

$$\mathbb{E}(X_T) = \mathbb{E}(X_S)$$

and since 0 is a allowed stopping time

$$\mathbb{E}(X_t) = \mathbb{E}(X_0)$$

for any bounded stopping time T .

If we think of X_n as the wealth at time n , and time T as the time we stop gambling then this says that no matter how we choose T , as long as it is bounded, then the expected wealth at the end is the same as what we started with. Simply put, the a gambler cannot gain an advantage by choosing when to stop playing based on information up to that point.

Example 21.1. Consider the stopping time

$$T = \inf\{n : X_n = 1\}$$

and take X s.t. $\mathbb{E}(X_0) = 0$. Then we have that $\mathbb{E}(X_t) = 1$. So this means T is not a bounded stopping time.

Theorem 6. The following are equivalent:

- (i) $(X_n)_{n \geq 0}$ is a super-martingale
- (ii) For any bounded stopping times T and any stopping time S ,

$$\mathbb{E}(X_T | \mathcal{F}_S) \leq X_{S \wedge T}$$

- (iii) (X_n^T) is a super-martingale for any stopping time T
- (iv) For bounded stopping times S, T s.t. $S \leq T$, we have

$$\mathbb{E}X_T \leq \mathbb{E}X_S$$

Proof

- (ii) $\rightarrow$ (iii) Consider $(X_n^{T'})_{n \geq 0}$ for a stopping time T' . To check if this is a super-martingale, we need to prove that whenever $m \leq n$,

$$\mathbb{E}(X_{n \wedge T'} | \mathcal{F}_m) \leq X_{m \wedge T'}$$

but this is given by (ii) by taking $S = m$ and $T = T' \wedge n$

- (ii) $\rightarrow$ (iv) by the tower law

- (iii) $\rightarrow$ (i) let $T = \infty$
- (iv) $\rightarrow$ (ii) obvious by construction
- (i) $\rightarrow$ (ii) Assume $T \leq n$. Then

$$\begin{aligned} X_T &= X_{S \wedge T} + \sum_{S \leq k < T} (X_{k+1} - X_k) \\ &= X_{S \wedge T} + \sum_{k=0}^n (X_{k+1} - X_k) \mathbb{1}_{S \leq k < T} \end{aligned}$$

Note $\{S \leq k < T\} = \{S \leq k\} \cap \{T \leq k\}^c \in \mathcal{F}_k$. Let $A \in \mathcal{F}_S$. Then $A \cap \{S \leq k\} \in \mathcal{F}_k$ by definition of $\mathcal{F}_S$. So $A \cap \{S \leq k < T\} \in \mathcal{F}_k$. Apply $\mathbb{E}$ to $(*) \times \mathbb{1}_A$. Then we have

$$\mathbb{E}(X_T \mathbb{1}_A) = \mathbb{E}(X_{S \wedge T} \mathbb{1}_A) + \sum_{k=0}^n \mathbb{E}(X_{k+1} - X_k) \mathbb{1}_{A \cap \{S \leq k < T\}}$$

But for all k , we know

$$\mathbb{E}(X_{k+1} - X_k) \mathbb{1}_{A \cap \{S \leq k < T\}} \leq 0$$

since X is a super-martingale. So it follows that for all $A \in \mathcal{F}_S$, we have

$$\mathbb{E}(X_T \mathbb{1}_A) \leq \mathbb{E}(X_{S \wedge T} \mathbb{1}_A)$$

but since $X_{S \wedge T}$ is $\mathcal{F}_{S \wedge T}$ measurable, it is in particular $\mathcal{F}_S$ measurable. So it follows that for all $A \in \mathcal{F}_S$, we have

$$\mathbb{E}(\mathbb{E}(X_T | \mathcal{F}_S) \mathbb{1}_A) \leq \mathbb{E}(X_{S \wedge T} \mathbb{1}_A)$$

- (iv) $\rightarrow$ (i) fix $m \leq n$ and $A \in \mathcal{F}_m$. Take

$$T = m \mathbb{1}_A + X_n \mathbb{1}_{A^c}$$

So we have

$$\begin{aligned} 0 &\geq \mathbb{E}(X_n) - \mathbb{E}(X_T) \\ &= \mathbb{E}(X_n) - \mathbb{E}(X_n \mathbb{1}_{A^c}) - \mathbb{E}(X_m \mathbb{1}_A) = \mathbb{E}(X_n \mathbb{1}_A) - \mathbb{E}(X_m \mathbb{1}_A) \end{aligned}$$

2.3 Martingale convergence theorems

One nice property of martingales is that they have nice convergence properties. We will demonstrate them by introducing the concept of an upcrossing.

Theorem 7. (Almost sure martingale convergence theorem). Suppose $(X_n)_{n \geq 0}$ is a super-martingale that is bounded in L^1 ($\sup_n \mathbb{E}|X_n| < \infty$). Then there exists an $\mathcal{F}_\infty$ -measurable $X_\infty \in L^1$ such that

$$X_n \rightarrow X_\infty \text{ a.s. as } n \rightarrow \infty$$

To begin, we'll need a convenient characterization of when a series converges.

Definition 22. (Upcrossing). Let (x_n) be a sequence and (a, b) an interval. An *upcrossing* of (a, b) by (x_n) is a sequence $j, j+1, \dots, k$ s.t. $x_j \leq a$ and $x_k \geq b$. We define

$$U_n[a, b, (x_n)] = \text{number of disjoint upcrossings contained in } \{1, \dots, n\}$$

$$U[a, b, (x_n)] = \lim_{n \rightarrow \infty} U_n[a, b, x]$$

Lemma 5. Let $(x_n)_{n \geq 0}$ be a sequence of numbers. Then x_n converges in $\mathbb{R}$ iff

- $\liminf |x_n| < \infty$
- For all $a, b \in \mathbb{Q}$ with $a < b$, we have $U[a, b, (x_n)] < \infty$

For our martingales, since they are bounded in L^1 , Fatou's lemma tells us that $\mathbb{E} \liminf |X_n| < \infty \rightarrow \liminf |X_n| = 0$ a.s. Thus, it remains to show that for any $a < b \in \mathbb{Q}$, we have $\mathbb{P}(U[a, b, (X_n)] = \infty) = 0$. This is a consequence of Doob's upcrossing lemma.

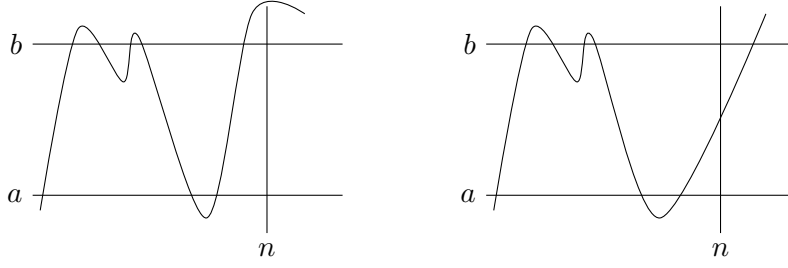
Lemma 6. (Doob's upcrossing lemma). If X_n is a super-martingale, then

$$(b - a)\mathbb{E}(U_n[a, b(X_n)]) \leq \mathbb{E}(X_n - a)^-$$

Proof Assume that X is a positive definite super-martingale. We define stopping times S_k, T_k as follows:

- $T_0 = 0$
- $S_{k+1} = \inf\{n : X_n \leq a, n \geq T_n\}$
- $T_{k+1} = \inf\{n : X_n \geq b, n \geq S_{k+1}\}$

Given an n , we want to count the number of upcrossings before n , there are two cases we'd like to distinguish:



Now consider the sum

$$\sum_{k=1}^n X_{T_k \wedge n} - X_{S_k \wedge n}.$$

In the first case, this is equal to

$$\sum_{k=1}^{U_n} X_{T_k} - X_{S_k} + \sum_{k=U_n+1}^n X_n - X_n \geq (b-a)U_n$$

In the second case, it is equal to

$$\sum_{k=1}^{U_n} X_{T_k} - X_{S_k} + (X_n - X_{S_{U_n+1}}) + \sum_{k=U_n+2}^n X_n - X_n \geq (b-a)U_n + (X_n - X_{S_{U_n+1}})$$

Thus, in general, we have

$$\sum_{k=1}^n X_{T_k \wedge n} - X_{S_k \wedge n} \geq (b-a)U_n + (X_n - X_{S_{U_n+1} \wedge n})$$

By definition, $S_k < T_k \leq n$. So the expectation of the LHS is always non-negative by super-martingale convergence, and thus

$$0 \geq (b-a)\mathbb{E}U_n + \mathbb{E}(X_n - X_{S_{U_n+1} \wedge n})$$

Then observe that

$$X_n - X_{S_{U_n+1}} \geq -(X_n - a)^-$$

The almost sure martingale convergence theorem is nice, but it often doesn't give enough results for dimensions higher than 1. For $p > 1$, if we want convergence in L^p , we need the sequence to be at least L^p bounded. However for $p = 1$ we will need a bit more than being bounded in L^1 , we will need uniform integrability.

Lemma 7. (Maximal inequality). Let (X_n) be a sub-martingale that is non-negative, or a martingale. Let

$$X_n^* = \sup_{k \leq n} |X_k|, \quad X^* = \lim_{n \rightarrow \infty} X_n^*$$

If $\lambda \geq 0$ then

$$\lambda \mathbb{P}(X_n^* \geq \lambda) \leq \mathbb{E}[|X_n| \mathbb{1}_{X_n^* \geq \lambda}]$$

In particular, we have

$$\lambda \mathbb{P}(X_n^* \geq \lambda) \leq \mathbb{E}[|X_n|]$$

Proof If X_n is a martingale, then $|X_n|$ is a sub-martingale. So it suffices to consider the case of a non-negative sub-martingale. We define the stopping time

$$T = \inf\{n : X_n \geq \lambda\}$$

By optional stopping,

$$\begin{aligned} \mathbb{E}X_n &\geq \mathbb{E}X_{T \wedge n} \\ &= \mathbb{E}X_T \mathbb{1}_{T \leq n} + \mathbb{E}X_n \mathbb{1}_{T > n} \\ &\geq \lambda \mathbb{P}(T \leq n) + \mathbb{E}X_n \mathbb{1}_{T > n} \\ &= \lambda \mathbb{P}(X_n^* \geq \lambda) + \mathbb{E}X_n \mathbb{1}_{T > n} \end{aligned}$$

Lemma 8. (Doob's L^p inequality). For $p > 1$, we have

$$\|X_n^*\|_p \leq \frac{p}{p-1} \|X_n\|_p$$

for all n

Proof Let $k > 0$, and consider

$$\|X_n^* \wedge k\|_p^p = \mathbb{E}|X_n^* \wedge k|^p$$

We use the fact that

$$x^p \int_0^x ps^{p-1} ds$$

So we have

$$\begin{aligned}
\|X_n^* \wedge k\|_p^p &= \mathbb{E}|X_n^* \wedge k|^p \\
&= \mathbb{E} \int_0^{X_n^* \wedge k} px^{p-1} dx \\
&= \mathbb{E} \int_0^k px^{p-1} \mathbf{1}_{X_n^* \geq x} dx \\
&= \int_0^k px^{p-1} \mathbb{P}(X_n^* \geq x) dx && \text{(Fubini's)} \\
&\leq \int_0^k px^{p-2} \mathbb{E} X_n \mathbf{1}_{X_n^* \geq x} dx && \text{(Maximal inequality)} \\
&= \mathbb{E} X_n \int_0^k px^{p-2} \mathbf{1}_{X_n^* \geq x} dx && \text{(Fubini's)} \\
&= \frac{p}{p-1} \mathbb{E} X_n (X_n^* \wedge k)^{p-1} \\
&\leq \frac{p}{p-1} \|X_n\|_p (\mathbb{E}(X_n^* \wedge k)^p)^{\frac{p-1}{p}} && \text{(Holder's)} \\
&= \frac{p}{p-1} \|X_n\|_p \|X_n^* \wedge k\|_p^{p-1}
\end{aligned}$$

Then take the limit $k \rightarrow \infty$ and divide by $\|X_n^*\|_p^{p-1}$.

Theorem 8. (L^p martingale convergence theorem). Let $(X_n)_{n \geq 0}$ be a martingale, and $p > 1$. Then the following are equivalent:

- (i) $(X_n)_{n \geq 0}$ is bounded in L^p , i.e. $\sup_n \mathbb{E}|X_n|^p < \infty$
- (ii) $(X_n)_{n \geq 0}$ converges as $n \rightarrow \infty$ to a random variable $X_\infty \in L^p$ a.s. and in L^p
- (iii) There exists a random variable $Z \in L^p$ s.t.

$$X_n = \mathbb{E}(Z | \mathcal{F}_n)$$

Moreover, in (iii), we always have $X_\infty = \mathbb{E}(Z | \mathcal{F}_\infty)$

This gives a bijection between martingales bounded in L^p and $L^p(\mathcal{F}_\infty)$, sending $(X_n)_{n \geq 0} \rightarrow X_\infty$

Proof

- (i) $\rightarrow$ (ii): If $(X_n)_{n \geq 0}$ is bounded in L^p , then it is bounded in L^1 . So by the martingale convergence theorem, we know $(X_n)_{n \geq 0}$ converges almost surely to X_∞ . By Fatou's lemma, we have $X_\infty \in L^p$.

Now by monotone convergence, we have

$$\|X^*\|_p = \lim_n \|X_n^*\|_p \leq \frac{p}{p-1} \sup_n \|X_n\|_p < \infty$$

By the triangle inequality, we have

$$|X_n - X_\infty| \leq 2X^* \text{ a.s.}$$

So by dominated convergence, we know that $X_n \rightarrow X_\infty$ in L^p

- (ii) $\rightarrow$ (iii): Take $Z = X_\infty$. We want to prove that

$$X_m = \mathbb{E}(X_\infty | \mathcal{F}_m)$$

To do so, we show that $\|X_m - \mathbb{E}(X_\infty | \mathcal{F}_m)\|_p = 0$. For $n \geq m$, we know this is equal to

$$\|E(X_n | \mathcal{F}_m) - \mathbb{E}(X_\infty | \mathcal{F}_m)\|_p = \|\mathbb{E}(X_n - X_\infty | \mathcal{F}_m)\|_p \leq \|X_n - X_\infty\|_p \rightarrow 0$$

as $n \rightarrow \infty$, where the last step uses Jensen's. But it is also a constant. So we are done.

- (iii) $\rightarrow$ (i): Since expectation decreases L^p norms, we already know that $(X_n)_{n \geq 0}$ is L^p -bounded.

$$\mathbb{E}X_\infty \mathbb{1}_A = \mathbb{E}(\mathbb{E}(Z | \mathcal{F}_\infty) \mathbb{1}_A)$$

But if $A \in \mathcal{F}_N$, then

$$\begin{aligned} \mathbb{E}X_\infty \mathbb{1}_A &= \lim_{n \rightarrow \infty} \mathbb{E}X_n \mathbb{1}_A \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(\mathbb{E}(Z | \mathcal{F}_n) \mathbb{1}_A) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(\mathbb{E}(Z | \mathcal{F}_\infty) \mathbb{1}_A), \end{aligned}$$

since $\mathbb{1}_A$ is $\mathcal{F}_n$ -measurable.

We can now finish the $p = 1$ case with uniform integrability.

Theorem 9 (Convergence in L^1). Let $(X_n)_{n \geq 0}$ be a martingale. Then the following are equivalent:

1. $(X_n)_{n \geq 0}$ is uniformly integrable.
2. $(X_n)_{n \geq 0}$ converges almost surely and in L^1 .
3. There exists $Z \in L^1$ such that $X_n = \mathbb{E}(Z | \mathcal{F}_n)$ almost surely.

Moreover, $X_\infty = \mathbb{E}(Z | \mathcal{F}_\infty)$.

Proof

- (i) $\rightarrow$ (ii): Let $(X_n)_{n \geq 0}$ be uniformly integrable. Then $(X_n)_{n \geq 0}$ is bounded in L^1 . So the $(X_n)_{n \geq 0}$ converges to X_∞ almost surely. Uniform integrability implies that in fact $X_n \rightarrow X_\infty$ in L^1 .

- (ii) $\rightarrow$ (iii): Same as the L^p case.
- (iii) $\rightarrow$ (i): For any $Z \in L^1$, the collection $\mathbb{E}(Z|\mathcal{G})$ ranging over all σ -subalgebras $\mathcal{G}$ is uniformly integrable.

Thus, there is a bijection between uniformly integrable martingales and $L^1(\mathcal{F}_\infty)$.

Theorem 10. If $(X_n)_{n \geq 0}$ is a uniformly integrable martingale, and S, T are arbitrary stopping times, then $\mathbb{E}(X_T|\mathcal{F}_S) = X_{S \wedge T}$. In particular $\mathbb{E}X_T = X_0$.

Note that we now allow arbitrary stopping times, so T may be infinite with non-zero probability. Hence we define

$$X_T = \sum_{n=0}^{\infty} X_n \mathbb{1}_{T=n} + X_\infty \mathbb{1}_{T=\infty}$$

Proof By optional stopping, for every n , we know that

$$\mathbb{E}(X_{T \wedge n}|\mathcal{F}_S) = X_{S \wedge T \wedge n}$$

We want to be able to take the limit as $n \rightarrow \infty$. To do so, we need to show that things are uniformly integrable.

$$\begin{aligned} X_{T \wedge n} &= \mathbb{E}(X_n|\mathcal{F}_{T \wedge n}) \\ &= \mathbb{E}(\mathbb{E}(X_\infty|\mathcal{F}_n)|\mathcal{F}_{T \wedge n}) \\ &= \mathbb{E}(X_\infty|\mathcal{F}_{T \wedge n}). \end{aligned}$$

So we know $(X_n^T)_{n \geq 0}$ is uniformly integrable, and hence $X_{n \wedge T} \rightarrow X_T$ almost surely and in L^1 .

$$\|\mathbb{E}(X_{n \wedge T} - X_T|\mathcal{F}_S)\|_1 \leq \|X_{n \wedge T} - X_T\|_1 \rightarrow 0 \text{ as } n \rightarrow \infty.$$

So it follows that $\mathbb{E}(X_{n \wedge T}|\mathcal{F}_S) \rightarrow \mathbb{E}(X_T|\mathcal{F}_S)$ as $n \rightarrow \infty$.

2.4 Uniformly Integrable Martingales

Definition 23. (Backwards filtration). A *backwards filtration* on a measurable space (E, Σ) is a sequence of σ -algebras $\hat{\mathcal{F}} \subseteq \Sigma$ s.t. $\hat{\mathcal{F}}_{n+1} \subseteq \hat{\mathcal{F}}_n$. We define

$$\hat{\mathcal{F}}_\infty = \bigcap_{n \geq 0} \hat{\mathcal{F}}_n$$

Theorem 11. Let $Y \in L^1$, and let $\hat{\mathcal{F}}_n$ be a backwards filtration, then

$$\mathbb{E}(Y|\hat{\mathcal{F}}_n) \rightarrow \mathbb{E}(Y|\hat{\mathcal{F}}_\infty)$$

a.s. in L^1 . A process in this form is known as a *backwards martingale*.

Proof We wts $\mathbb{E}(Y|\hat{\mathcal{F}}_n)$ converges, verify that it converges to $\mathbb{E}(Y|\hat{\mathcal{F}}_\infty)$.

Let $X_n = \mathbb{E}(Y|\hat{\mathcal{F}}_n)$. For all $n \geq 0$, the process $(X_{n-k})_{0 \leq k \leq n}$ is a martingale by the tower property, and so is $(-X_{n-k})_{0 \leq k \leq n}$. For all $a < b$, the number of upcrossings of $[a, b]$ by $(X_k)_{0 \leq k \leq n}$ equals the number of upcrossings of $[-b, -a]$ by $(-X_{n-k})_{0 \leq k \leq n}$.

Since X_n is a martingale, $X_n \rightarrow X_\infty$ a.s. and in L^1 for some X_∞ . We can justify that by showing

$$\begin{aligned} EX_\infty \mathbb{1}_A &= \lim_{n \rightarrow \infty} EX_n \mathbb{1}_A \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(\mathbb{E}(Y|\hat{\mathcal{F}}_\infty) \mathbb{1}_A) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(Y \mathbb{1}_A) \\ &= \mathbb{E}(Y \mathbb{1}_A) \\ &= \mathbb{E}(\mathbb{E}(Y|\hat{\mathcal{F}}_\infty) \mathbb{1}_A) \end{aligned}$$

Theorem 12. (Kolmogorov 0-1 law). Let $(X_n)_{n \geq 0}$ be independant RVs. Let

$$\hat{\mathcal{F}}_n = \sigma(X_{n+1}, X_{n+2}, \dots)$$

Then the *tail σ -algebra* $\hat{\mathcal{F}}_\infty$ is trivial, i.e. $\mathbb{P}(A) \in \{0, 1\} \forall A \in \hat{\mathcal{F}}_\infty$

Proof Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Then $\mathcal{F}_n$ and $\hat{\mathcal{F}}_n$ are independant. Then for all $A \in \hat{\mathcal{F}}_\infty$, we have

$$\mathbb{E}(\mathbb{1}_A | \mathcal{F}_n) = \mathbb{P}(A)$$

but the LHS is a martingale so it converges a.s. in L^1 to $\mathbb{E}(\mathbb{1}_A | \mathcal{F}_\infty)$. Since $\mathbb{1}_A$ is $\mathcal{F}_\infty$ -measurable ($\hat{\mathcal{F}}_\infty \subseteq \mathcal{F}_\infty$), $\mathbb{E}(\mathbb{1}_A | \mathcal{F}_\infty) = \mathbb{1}_A$. So $\mathbb{1}_A = \mathbb{P}(A)$ a.s.

Theorem 13. (Strong law of large numbers) Let $(X_n)_{n \geq 1}$ be iid RVs in L^1 , with $\mathbb{E}X_1 = \mu$. Let

$$S_n = \sum_{i=1}^n X_i$$

then $S_n/n \rightarrow \mu$ as $n \rightarrow \infty$ a.s. in L^1

Proof

$$\begin{aligned} S_n &= \mathbb{E}(X_n | S_n) \\ &= \sum_{i=1}^n \mathbb{E}(X_i | S_n) \\ &= n\mathbb{E}(X_1 | S_n) \end{aligned}$$

Define a backwards filtration to be:

$$\mathcal{F}_n = \sigma(S_n, S_{n+1}, \dots) = \sigma(S_n, X_{n+1}, X_{n+2}, \dots) = \sigma(S_n, \tau_n)$$

where $\tau_n = \sigma(X_{n+1}, X_{n+2}, \dots)$. Since τ_n is independent of $\sigma(X_1, S_n)$, we know

$$\frac{S_n}{n} = \mathbb{E}(X_1 | S_n) = \mathbb{E}(X_1 | \hat{\mathcal{F}}_n)$$

thus by backwards martingale convergence, we know

$$\frac{S_n}{n} \rightarrow \mathbb{E}(X_1 | \hat{\mathcal{F}}_\infty)$$

but by the kolmogorov 0-1 law, we know that $\hat{\mathcal{F}}_\infty$ is trivial. So we know that $\mathbb{E}(X_1 | \hat{\mathcal{F}}_\infty)$ is almost constant which means that $\mathbb{E}(\mathbb{E}(X_1 | \hat{\mathcal{F}}_\infty)) = \mathbb{E}(X_1) = \mu$.

If we have a measure space (E, Σ, μ) and $f \in m\Sigma^+$, then

$$\nu(A) = \mu(f \mathbb{1}_A)$$

is a measure on Σ . We say f is a density of ν wrt μ .

This means that ν is given by the density function f wrt μ , we have

$$0 = \nu(\{x\}) = \mu(f \mathbb{1}_{\{x\}}) = f(x)$$

So that means $f(x) = 0$, but that raises the problem that we cannot establish a lebesgue measure on this function. The problem with this is that μ is not a σ -finite measure.

Theorem 14. (Radon-Nikodym). Let $(\Omega, \mathcal{F})$ be a measurable space, and $\mathbb{Q}$ and $\mathbb{P}$ be two probability measures on $(\Omega, \mathcal{F})$. Then the following are equivalent:

- (i) $\mathbb{Q}$ is absolutely continuous wrt $\mathbb{P}$, i.e. for any $A \in \mathcal{F}$, if $\mathbb{P}(A) = 0$, then $\mathbb{Q}(A) = 0$
- (ii) For any $\epsilon > 0$, there exists $\delta > 0$ such that for all $A \in \mathcal{F}$, if $\mathbb{P}(A) \leq \delta$, then $\mathbb{Q}(A) \leq \epsilon$.
- (iii) There exists a random variable $X \geq 0$ such that

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}(X \mathbb{1}_A)$$

In this case, X is called the *Radon-Nikodym derivative* of $\mathbb{Q}$ with respect to $\mathbb{P}$ and we write it as

$$X = \frac{d\mathbb{Q}}{d\mathbb{P}}$$

Proof

- (iii) $\rightarrow$ (i) by construction
- (ii) $\rightarrow$ (iii) Let

$$\mathcal{F}_n = \sigma(F_1, F_2, \dots, F_n)$$

Since $\mathcal{F}_n$ is finite, we can write it as

$$\mathcal{F}_n = \sigma(A_{n,1}, \dots, A_{n,m_n})$$

where each $A_{n,i}$ is an atom i.e. if $B \not\subseteq A_{n,i}$ and $B \in \mathcal{F}_n$, then $B = \emptyset$. We define

$$X_n = \sum_{i=1}^{m_n} \frac{\mathbb{Q}(A_{n,i})}{\mathbb{P}(A_{n,i})} \mathbb{1}_{A_{n,i}}$$

This allows for any $A \in \mathcal{F}_n$ to have

$$\mathbb{E}_{\mathbb{P}}(X_n \mathbb{1}_A) = \mathbb{E}_{\mathbb{P}} \sum_{A_{n,i} \subseteq A} \frac{\mathbb{Q}(A_{n,i})}{\mathbb{P}(A_{n,i})} \mathbb{1}_{A_{n,i}} = \mathbb{Q}(A)$$

Thus if $A \in \mathcal{F}_n \subseteq \mathcal{F}_{n+1}$, we have

$$\mathbb{E}X_{n+1} \mathbb{1}_A = \mathbb{Q}(A) = \mathbb{E}X_n \mathbb{1}_A$$

So we know

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n) = X_n$$

Which means that $(X_n)_{n \geq 0}$ is adapted, so it is a martingale.

We next need to show that $(X_n)_{n \geq 0}$ is uniformly integrable which we can do with markov's inequality

$$\mathbb{P}(X_n \geq \lambda) \leq \frac{\mathbb{E}X_n}{\lambda} = \frac{1}{\lambda} \leq \delta$$

for λ large enough. Then

$$\mathbb{E}(X_n \mathbb{1}_{X_n \geq \lambda}) = \mathbb{Q}(X_n \geq \lambda) \leq \epsilon$$

Which shows uniform integrability. So we know $X_n \rightarrow X$ a.s. and in L^1 for some X . Then for all $A \in \bigcup_{n \geq 0} \mathcal{F}_n$, we have

$$\mathbb{Q}(A) = \lim_{n \rightarrow \infty} \mathbb{E}X_n \mathbb{1}_A = \mathbb{E}X \mathbb{1}_A$$

So $\mathbb{Q}(-)$ and $\mathbb{E}X \mathbb{1}_{(-)}$ agree on $\bigcup_{n \geq 0} \mathcal{F}_n$, which is a generating π -system for $\mathcal{F}$, so they must be the same.

- Suppose, for the sake of contradiction, that this wasn't true. Then there exists some $\epsilon > 0$ and some $A_1, A_2, \dots \in \mathcal{F}$ such that

$$\mathbb{Q}(A_n) \geq \epsilon, \quad \mathbb{P}(A_n) \leq \frac{1}{2^n}$$

Since $\sum_n \mathbb{P}(A_n)$ is finite, by borel cantelli

$$\mathbb{P} \limsup A_n = 0$$

but by the dominated convergence theorem

$$\begin{aligned} \mathbb{Q} \limsup A_n &= \mathbb{Q} \left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m \right) \\ &= \lim_{k \rightarrow \infty} \mathbb{Q} \left(\bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m \right) \\ &\geq \lim_{k \rightarrow \infty} \mathbb{Q} \left(\bigcup_{m=k}^{\infty} A_m \right) \\ &\geq \epsilon \end{aligned}$$

which arises a contradiction.

Definition 24. (Transition matrix). A transition matrix is a matrix $P = (p_{xy})_{x,y \in E}$ s.t. each $p_x = (p_{x,y})_{y \in E}$ is a probability measure on E

Definition 25. (Markov chain). An adapted process (X_n) is called a markov chain if for any n and $A \in \mathcal{F}_n$ such that $\{x_n = x\} \supseteq A$ we have

$$\mathbb{P}(X_{n+1} = y | A) = p_{xy}$$

3 Abracadabra

Non-Martingale - If a monkey is equally likely to type any of the 26 capital letters, how long on average will it take him to produce the sequence *ABRACADABRA*?

This question will be the focus of my combinatorial approach to the problem.

Martingale - At each of the times $1, 2, 3, \dots$, a monkey types a capital letter at random, the sequence of letters typed forming an IID sequence of RVs each chosen uniformly from amongst the 26 possible capital letters.

Just before each time $n = 1, 2, \dots$, a new gambler arrives on the scene. He bets \$1 that

the n^{th} letter will be A

If he loses, he leaves. If he wins, he will receive \$26, all of which he bets on the event that

the $(n + 1)^{th}$ letter will be B

If he loses, he leaves. If he wins, he bet his entire fortune of $\$26^2$ that

the $(n + 2)^{th}$ letter will be R

and so on through the ABRACADABRA sequence. Let T be the first time by which the monkey has produced the consecutive sequence ABRACADABRA. Explain why martingale theory makes it intuitively obvious that

$$\mathbb{E}(T) = 26^{11} + 26^4 + 26$$

3.1 Combinatorial structures

A combinatorial class $\mathcal{A}$ is a finite set where a size function is defined with the following properties

1. the size of an element is a non negative number
2. the number of elements of any given size is finite

Definition 26. Two combinatorial classes $\mathcal{A}$ and $\mathcal{B}$ are said to be (combinatorially) isomorphic ($\mathcal{A} \approx \mathcal{B}$), iff their counting sequences are identical. Another expression for this is there exists a bijection from $\mathcal{A}$ to $\mathcal{B}$ that preserves size.

The ordinary generating function (OGF) of a sequence A_n is the series:

$$A(z) = \sum_{n=0}^{\infty} A_n z^n$$

where z marks the size in the generating function

Generating functions allow for the reduction of a lot of problems down to combinatorial, for example:

$$\begin{array}{ccccccc} \mathcal{H} = & \begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} & \begin{array}{c} \bullet \text{---} \bullet \end{array} & \begin{array}{c} \bullet \quad \bullet \\ \diagdown \quad \diagup \\ \bullet \end{array} & \begin{array}{c} \bullet \quad \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} & \bullet & \begin{array}{c} \bullet \\ | \\ \bullet \text{---} \bullet \end{array} & \begin{array}{c} \bullet \text{---} \bullet \text{---} \bullet \end{array} \\ & zzzz & zz & zzz & zzzz & z & zzzz & zzz \\ & + z^4 & + z^2 & + z^3 & + z^4 & + z & + z^4 & + z^3 \\ H(z) = & & & & z + z^2 + 2z^3 + 3z^4 & & & \end{array}$$

where size is taken as the number of vertices. Each vertex is replaced by z which allows us to separate from the graph structure. The result is a generating function corresponding to each graph by gathering all the terms. Note: This is not the only way to generate a OGF, for example picking number of edges would produce: $1 + z + z^2 + 2z^3 + z^4 + z^6$, thus our selection of z will heavily impact what our OGF represents, and can be used to solve.

Unrelated to this question, but bivariate generating functions can be generated if you define each parameter as it's own variable. For example treating vertices as z and edges as y will produce: $H(z, y) = z + z^2y + z^3y^2 + z^3y^3 + z^4y^3 + z^4y^4 + z^4y^6$ as each pairwise z and y are multiplied. These have their own results worth looking into. In essence, this is the combinatorial version of a joint distribution on $f(z, y)$.

Lemma 9. (Coefficient extraction) We generally let $[z^n]f(z)$ denote the operation of extracting the coefficient of z^n in the power series $f(z) = \sum f_n z^n$, so that

$$[z^n] \left(\sum_{n \geq 0} f_n z^n \right) = f_n$$

Definition 27. Let Φ be a m -ary construction that associates to any collection of classes $\Phi[\mathcal{B}^{(1)}, \dots, \mathcal{B}^{(m)}]$ a new class

$$\mathcal{A} = \Phi[\mathcal{B}^{(1)}, \dots, \mathcal{B}^{(m)}]$$

The construction Φ is admissible iff the counting sequence (A_n) of $\mathcal{A}$ only depends on the counting sequences $(B_n^{(1)}), \dots, (B_n^{(m)})$ of $\mathcal{B}^{(1)}, \dots, \mathcal{B}^{(m)}$.

Theorem 15. (Basic admissibility, unlabelled universe). The constructions of union, cartesian product, sequence, powerset, multiset, and cycle are all admissible. The associated operators are:

Sum	$\mathcal{A} = \mathcal{B} + \mathcal{C}$	$\implies$	$A(z) = B(z) + C(z)$
Product	$\mathcal{A} = \mathcal{B} \times \mathcal{C}$	$\implies$	$A(z) = B(z) * C(z)$
Sequence	$\mathcal{A} = SEQ(\mathcal{B})$	$\implies$	$A(z) = \frac{1}{1-B(z)}$
Powerset	$\mathcal{A} = PSET(\mathcal{B})$	$\implies$	$A(z) = \left\{ \prod_{n \geq 1} (1 + z^n)^{B_n} \right. \\ \left. exp(\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} B(z^k)) \right.$
Multiset	$\mathcal{A} = MSET(\mathcal{B})$	$\implies$	$A(z) = \left\{ \prod_{n \geq 1} (1 - z^n)^{-B_n} \right. \\ \left. exp(\sum_{k=1}^{\infty} \frac{1}{k} B(z^k)) \right.$
Cycle	$\mathcal{A} = CYC(\mathcal{B})$	$\implies$	$A(z) = \sum_{k=1}^{\infty} \frac{\phi(k)}{k} \log \frac{1}{1-B(z^k)}$

Lemma 10. (Sequence construction) Admissibility for $\mathcal{A} = SEQ(\mathcal{B})$ (with $\mathcal{B}_0 = \emptyset$) follows from the union and product relations where

$$\mathcal{A} = \{\epsilon\} + \mathcal{B} + (\mathcal{B} \times \mathcal{B}) + (\mathcal{B} \times \mathcal{B} \times \mathcal{B}) + \dots,$$

so that

$$A(z) = 1 + B(z) + B(z)^2 + B(z)^3 + \dots = \frac{1}{1-B(z)}$$

where the geometric sum converges as a power series since the first term must be equal to 0 by our assumption.

3.2 Words and language

A finite alphabet $\mathcal{A}$ is composed of letters of size 1 (atomic). A *word* (or string in CS, sequence in Bio) is a finite sequence of letters without any separators. The set of all words is $\mathcal{W}$. We say that any subset of $\mathcal{W}$ is a language (not necessarily a natural language).

From the definition of the set of words $\mathcal{W}$, it follows that

$$\mathcal{W} \approx SEQ(\mathcal{A}) \implies W(z) = \frac{1}{1-mz}$$

where m is the cardinality of the alphabet. The generating function gives us the counting result that $W_n = m^n$.

Thought experiment: In a given word (using all capital letters), we want to find the

Definition 28. (Longest run) $a^{<k}$ is defined to be the collection of all words formed from a specific letter whose length is between 0 and $k - 1$. The OGF for a longest run

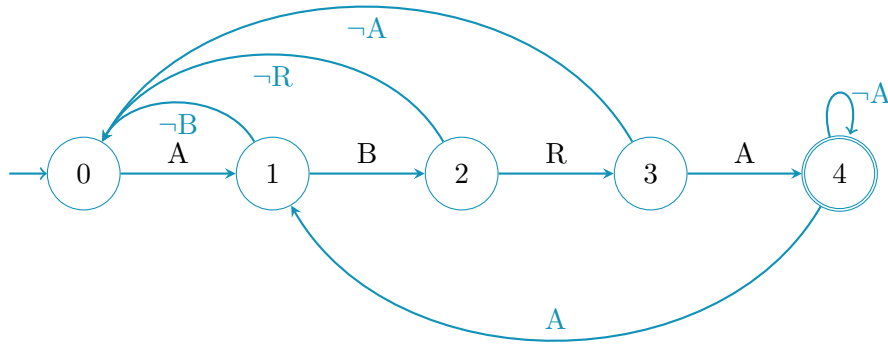
$$\mathcal{W}^{<k} = a^{<k}SEQ(ma^{<k}) \implies W^{<k}(z) = \frac{1 - z^k}{1 - z} \frac{1}{1 - z \frac{1 - z^k}{1 - z}} = \frac{1 - z^k}{1 - 2z + z^{k+1}}$$

The case of longest runs demonstrates the fact that nested constructions in sequences can be solved using combinatorial methods.

A language $\mathcal{L}$ is said to be S-regular ("specification-regular") if there exists a class $\mathcal{M}$ described by a regular specification such that $\mathcal{L}$ and $\mathcal{M}$ are combinatorially isomorphic $\mathcal{L} \approx \mathcal{M}$

Definition 29. (Finite automata) A finite automaton is a directed multigraph whose edges are labelled by letters of the alphabet $\mathcal{A}$. The vertices are referred to as states, and Q is the set of associated states. The initial state is $q_0 \in Q$, and the set of final states is $\bar{Q} \subseteq Q$

A word $w = w_1 \cdots w_n$ is accepted by the finite automaton if there exists a path in the multigraph connecting the initial state q_0 to one of the final states of $\bar{Q}$ and whose sequence of edge labels is precisely $w_1, \cdots, w_n$. A finite automaton holds only a finite memory and is essentially the combinatorial version of a *Markov chain* (see 2.4).



A language is said to be A -regular (automaton regular) if it coincides with the set of words accepted by the deterministic finite automaton. A language is S -regular iff it's A -regular.

Example 29.1. Words containing or excluding a pattern. Let a pattern $p = p_1 p_2 \cdots p_k$ and let $\mathcal{L}$ be the language of words containing at least one occurrence of p as a factor. Automamta theory implies that the set of words containing a pattern as a factor is A -regular. As a consequence: the OGF of the language of words containing a given factor pattern of length k is a rational function of degree at most $k + 1$. The automaton construction only provides the OGF in determinantal form, so the relationship between the rational form and structure of the pattern is not clear.

Definition 30. (Autocorrelation) To prevent this problem, we establish the notion of autocorrelation. For a given pattern p , there is a vector of bits $c = (c_0, \cdots, c_{k-1})$ that is defined to be

$$c_i = \mathbb{1}_{p_{i+1}p_{i+2}\cdots p_k = p_1p_2\cdots p_{k-i}}$$

In other words, the bit c_i is determined by shifting p right by i positions and putting a 1 if the remaining letters match the original p .

For example if $p = aabbaa$, then

0	a	a	b	b	a	a		
0	a	a	b	b	a	a		1
1		a	a	b	b	a		a
2			a	a	b	b		a
3				a	a	b		b
4					a	a		b
5						a		a

At $n = 0$, $aabbaa = aabbaa$. At $n = 4$, $aabb = bbaa$. At $n = 5$, $aabba = abbaa$.

The autocorrelation vector is then $c = (1, 0, 0, 0, 1, 1)$. The autocorrelation polynomial is then defined to be

$$c(z) := \sum_{j=0}^{k-1} c_j z^j$$

For the example pattern, this gives $c(z) = 1 + z^4 + z^5$.

Example 30.1. Let $S(z)$ be the OGF for words without the pattern p . We can build the function using a recurrence relation:

- Start with an empty word (1)
- Append any symbol mz to a valid word $S(z)$
- Subtract cases where appending the letter completes p , $(z^m/c(z))$

$$\begin{aligned}
S(z) &= 1 + mzS(z) - \frac{z^m}{c(z)}S(z) \\
S(z) - mzS(z) + \frac{z^m}{c(z)}S(z) &= 1 \\
S(z)(1 - mz + \frac{z^m}{c(z)}) &= 1 \\
S(z) &= \frac{1}{1 - mz + \frac{z^m}{c(z)}} \\
S(z) &= \frac{c(z)}{c(z) - c(z)mz + z^m} \\
S(z) &= \frac{c(z)}{c(z)(1 - mz) + z^m}
\end{aligned}$$

Example 30.2. Using the tools we have built up, let's approach the abracadabra problem using a finite automata. We are given the word *ABRACADABRA*, and we want to find the expected waiting time to produce the sequence. Let's start by finding the number of strings which do not contain the patterns within the word.

$$\begin{aligned}
c &= [[p_{i+1}p_{i+2}\dots p_k = p_1p_2\dots p_{k-i}]] \\
c &= (1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1) \\
c(z) &= 1 + z^7 + z^{10} \\
S(z) &= \frac{1 + z^7 + z^{10}}{z^{11} + (1 - 26z)(1 + z^7 + z^{10})}
\end{aligned}$$

The average wait time for the first occurrence of the pattern in a random string is $S(1/m) = m^k c(1/m)$

(The Probability link) - If $S(z) = \sum_n s_n z^n$, then s_n would be the number of strings of length n without the pattern. So s_n/m^n is the probability that the string does not contain the pattern $s_n/m^n = P(X > n)$ where X is the position of the first occurrence of the pattern. Essentially, we built a probability measure revolving around our OGF. Notably, OGFs and probability measures have a lot in common, and OGFs give the exact values but requires a lot more information about the sequence. This question was phrased so a martingale would be a perfect fit. By finding the number of strings without the pattern, we establish a stopping time for the sequence in which we can use it as a probability measure which corresponds the the expected value wait time until the sequence produces the desired string.

For the pattern ABRACADABRA with an alphabet of $m = 26$ letters, the average

wait time for the first occurrence should be equal to $S(1/m)$

$$\begin{aligned}
 S(1/26) &= \frac{1 + (1/26)^7 + (1/26)^{10}}{(1/26)^{11} + (1 - 26(1/26))(1 + (1/26)^7 + (1/26)^{10})} \\
 &= \frac{1 + 1/26^7 + 1/26^{10}}{1/26^{11}} \\
 &= \frac{1}{1/26^{11}} + \frac{1/26^7}{1/26^{11}} + \frac{1/26^{10}}{1/26^{11}} \\
 &= 26^{11} + 26^4 + 26
 \end{aligned}$$

3.3 Martingale approach

First and most important to note is that successive trials in this problem is not independent (something that I fell for initially). At each stage of the game the person has a $1/26$ chance of winning 26 times what he bets. Let's assume that the monkey typed everything correctly and find the maximum amount of money that can be won. Let's take the first gambler to be our winner who took home the $\$26^{11}$ at time 11. There can be no more winning gamblers until the 4th A because if the first gambler won the $\$26^{11}$, then the 2nd, 3rd, $\dots$ 7th gamblers all must have lost. The gambler who started winning at the 4th A can win at maximum $\$26^4$ because there are four more letters that the monkey can type correctly. Lastly the 11th gambler can also win $\$26$ because the last letter is an A.

X = not a hit

Lucky Gambler: ABRACADABRA

Semi Lucky Gambler: XXXXXXXXABRA

Last Minute Gambler: XXXXXXXXXXXXA

Any gambler who won in the middle doesn't count towards the problem as the lost their earnings on the next play.

Doob's Optional stopping lemma states

$$\mathbb{E}X_T = \mathbb{E}X_0$$

thus $\mathbb{E}X_0 = 0$ because nothing happens at time 0. The total winnings should be $\$26^{11} + \$26^4 + \$26$.

$$\begin{aligned}
 \mathbb{E}X_T &= \mathbb{E}(26^{11} + 26^4 + 26 - T) \\
 &= 26^{11} + 26^4 + 26 - \mathbb{E}(T) = 0 \\
 \iff E(T) &= 26^{11} + 26^4 + 26
 \end{aligned}$$